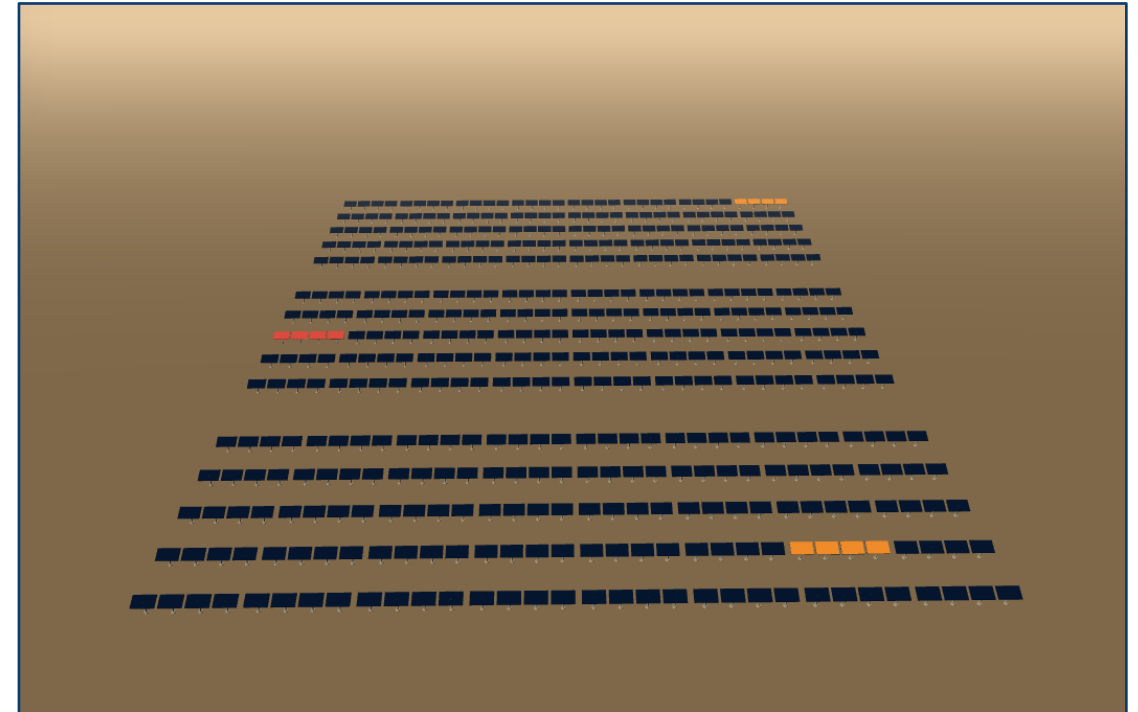


The plan, not the picture.

An AI agent that watches a 500 MW block of Bhadla Solar Park, sends a drone to verify what telemetry cannot, and hands the operator a ranked repair plan with a computed deadline. A person approves it before anything is scheduled.

Event	JSS AI FORGE 36, Round 1 idea submission
Track	AI for Industry 4.0: predictive maintenance, automation, digital twins
Team	SIGMOID
Members	Rehaan Ahmad Khan, Shantanu Singh, Lakshita Rawat, Krishna Agarwal
Prototype	Built and running. github.com/RAK2315/solar-proj



The running prototype: a 3D twin of 120 arrays. B-17 is the red one.

364 MW

delivered from 500 MW nameplate,
at 62.8 °C cell temperature

-41.7 %

on array B-17: 5 of its 7 strings
bypassed

3.07 MWh

lost over 72 h if nobody acts before
14:00

0.995

AP@50 for cracked panels, held-out
test split

THE PROBLEM: the plant knows its output fell, not why, or how long it can wait.

PLANT TELEMETRY (SCADA)

WHAT IT SAYS

An inverter or a string is producing less than it should

WHAT IT CANNOT SAY

Which module? Dirt or damage?
How urgent?

SO THE FAULT WAITS Someone drives out to look, or it sits until the next aerial survey. In our model, soiling and a cracked cell look the same from telemetry: only imaging separates them.

Annual

the usual cadence of aerial infrared inspection

Weeks to months

faults below the inverter go undetected

-41.7 %

one cracked array in our model: 3.07 MWh, ₹7,509 lost in 72 h

Sources: Turbine Logic and EPRI, ost.gov/servlets/purl/1960134; third figure, our PV model.

WHO BENEFITS



O&M crews
Which array, what repair, by when



Asset owners
Loss in MWh and in rupees



Grid off-taker
More contracted energy delivered

EXISTING GAPS

Approach	What it gives	What is still missing
Plant SCADA and inverter monitoring	Output fell, at an inverter or a string	Which module, why, and how urgent
Annual aerial infrared survey	A thermal map of the whole field	Flown once a year, so faults sit for weeks to months
Drone and AI inspection platforms (Raptor Maps, Zeitview, Sitemark)	Thermal imaging, AI defect classes, 3D twins under way	A report of defects. No deadline, no crew plan, no re-plan when conditions change
SURYA AGENT	A ranked work order with a computed deadline, re-derived live, arithmetic on screen	A prototype on simulated telemetry. Not yet connected to a real plant



What nobody hands over

A defect report tells you a panel is bad. It does not tell you when the damage stops being recoverable, which job the crew should do first, or what changes when the weather turns.



What we do not claim

The twin and the detection already exist elsewhere, and our thermal evidence comes from Raptor Maps' own open dataset. What we add is the step after the picture.

Gaps from our prior-art sweep of 4 Oct 2026, read from each company's public material. Not a benchmark.

PROPOSED SOLUTION: one closed loop, from a telemetry anomaly to an approved work order.



Actual problem Problems faced today

Faults wait for a yearly aerial survey

Telemetry cannot tell dirt from damage

A defect report carries no deadline

What to fix first is a judgement call

A plan goes stale when the weather turns

Automation nobody signed off

Proposed solution How we answer each one

Every array watched continuously against a physics model

A drone is sent only when imaging would add something

A deadline computed from the defect and the 72 h forecast

One fixed formula, with its arithmetic on screen

The queue and the crew day re-derive live

Nothing is scheduled until an operator approves

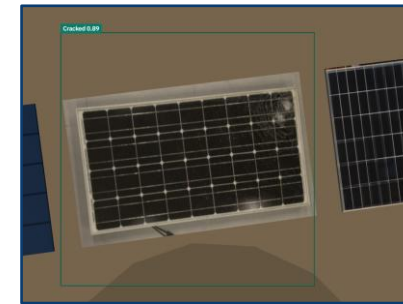
Honest wording: B-17 is diagnosed from a real thermal capture. The other 119 arrays are flagged from modelled signature. The build fails if the two are mixed.

The running prototype

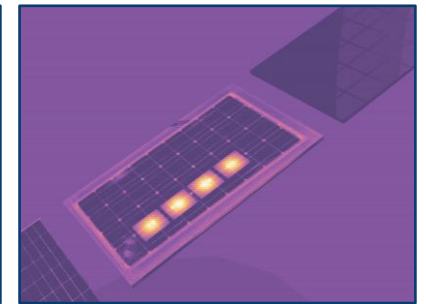
Not a mock-up. The detector we trained runs in the browser on the frame the drone's camera returned: Cracked 0.89 on B-17. The thermal pass beside it is a rendering of the simulated scene.



github.com/RAK2315/solar-proj



Detection, in the browser



Thermal pass, rendered

WHERE THE AI IS



Vision detector, trained by us
YOLOv8n on 921 labelled photographs. Cracked AP@50 0.995, held-out test split.



Prognosis that ends in an hour
A thermal-dose model and the 72 h forecast give "act before 14:00".



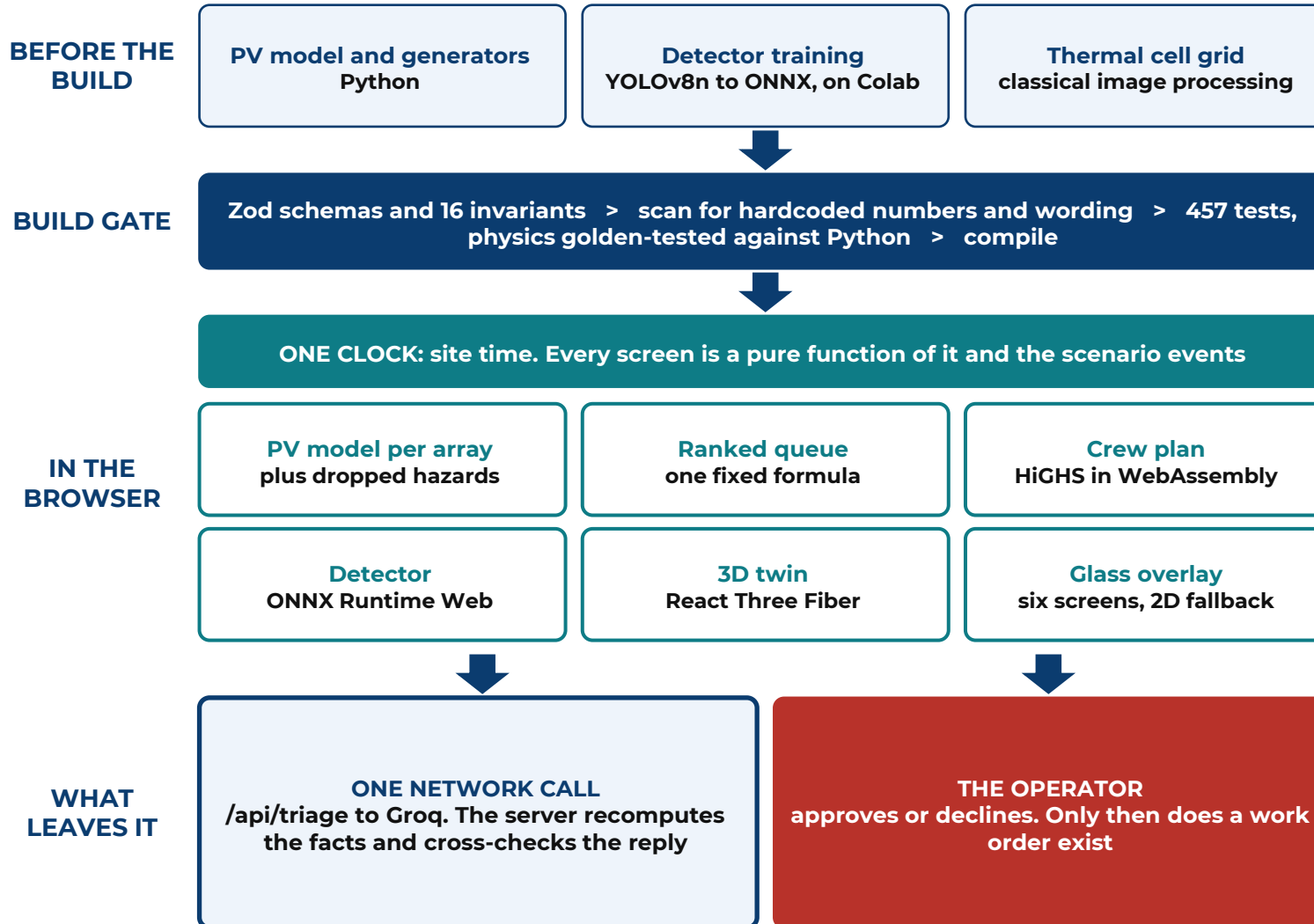
Agent reasoning
openai/gpt-oss-120b on Groq writes the triage. It never supplies a number.



Crew plan, exact optimisation
A mixed-integer program, HiGHS in WebAssembly, capped at 50 ms.

Not AI, on purpose: the queue order is a fixed formula. **In progress, no result claimed:** a thermal classifier on Raptor Maps' InfraredSolarModules.

A physics model, a trained detector and an exact solver, all running in the browser, behind a build that fails if a number drifts.



TECHNOLOGY STACK

-  **Frontend:** Next.js 15 + React 19 + TypeScript
-  **3D twin:** three.js + React Three Fiber
-  **Vision:** YOLOv8n on ONNX Runtime Web
-  **Optimiser:** HiGHS in WebAssembly
-  **Agent:** openai/gpt-oss-120b on Groq
-  **Offline:** Python + Colab T4 for training

WHY THESE CHOICES

-  **Everything in the browser** no GPU and no model server, so nothing can cold-start in front of a judge.
-  **An exact solver beside a heuristic** a new rule is one line of a model, and the answer is provable.
-  **The model writes words, not numbers** every figure comes from the physics and is cross-checked.
-  **One clock** seek backwards and every screen is correct. Same input, same site.

THE PLAN, NOT THE PICTURE: others detect and report. We hand over a deadline and a plan.

INNOVATION AND UNIQUENESS



A computed deadline

Not a flagged defect: an hour, worked out from the defect, its mechanism and the 72 h forecast.



A plan that re-derives live

Change the conditions and the queue, the deadlines and the crew day are worked out again.



Arithmetic on screen

The ranking formula with its inputs on every job, and a ? on every number for its source.



A human gate

The agent proposes. Only an operator's click creates a work order.



It says no

A dirty array gets a wash crew and no flight: imaging it would add nothing.

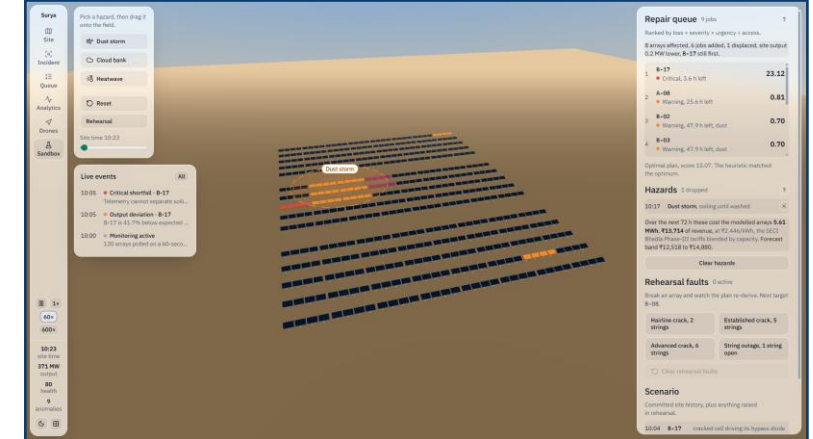
score = loss per day × severity × urgency ÷ access

B-17 as captured: 1.01 MWh × 3.0 × 7.69 ÷ 1.0 = 23.29

Urgency is 1 + 24 ÷ hours left. Next is A-08 at 0.81.

Rank	Job ID	Status	Loss per day	Severity	Urgency	Access	Score
1	B-17	Critical, 3.6 h left	1.01 MWh	3.0	7.69	1.0	23.29
2	A-08	Warning, 25.6 h left	0.28 MWh	1.5	1.94	1.0	0.81
3	C-31	Warning, 47.6 h left	0.28 MWh	1.5	1.50	1.4	0.45

Every job shows its working.



The what-if sandbox: a dust storm dropped on Zone B.

THE LIVE DEMONSTRATION

Name a hazard, drag it onto the field, and the whole plan re-derives.

What the screen said

8 arrays affected, 6 jobs added, 1 displaced, B-17 still first.

What it costs, in rupees

5.61 MWh, ₹13,714 over 72 h at ₹2.446/kWh. Band ₹12,518 to ₹14,880.

31 to 40 ms

from drop to the re-planned frame, in Chrome. Budget 150 ms. 60.1 fps.

FEASIBILITY



Technical: it is built. 457 tests on every build, 60.1 fps, a hazard re-planned in 31 to 40 ms, detector AP@50 0.995 on a held-out split.



Financial: no GPU and no model server. The detector and the solver run in the browser; the one outside call is to a language model.



Operational: one view in any modern browser. It is built to sit beside existing SCADA: the simulator is the one part to replace.

VIABILITY



Adoption: for O&M contractors and plant owners. Nothing to install, and every figure is in MWh and in rupees at the plant's own tariff.



Sustained: a build gate of 16 invariants fails the build, not the demo, if a number drifts. The same input always gives the same site.



Grows: every array is evaluated by the same function, so a larger plant is more rows, not a new design. Tested to 120 arrays.

POTENTIAL CHALLENGES AND RISKS

- The telemetry is simulated, not from a real plant. There is no live site behind the prototype.
- A language model can invent a number, or be rate-limited on the day. It is the one outside service we depend on.
- The detector learned from ground-level photographs. And we hold real thermal evidence for one array only.
- An assumption can pass for a fact. Hazard strengths, crew hours and the forecast band are ours.
- A weak laptop may not hold the 3D twin. A demo that stutters is worse than none.

HOW WE ADDRESSED THEM

- Every coefficient is stated and the equations are NREL PVWatts. The browser model is tested against the Python on every build.
- It writes words, never numbers. The server recomputes every fact and rejects a reply that disagrees. Without it, the rest still works.
- Reported per class on a held-out split, and scored on the drone's own frame: 0.89. Only B-17 is called diagnosed.
- Each is labelled on screen as an assumption. The thermal classifier is not built, so no score is claimed for it.
- A 2D map takes over by itself below 30 frames a second, on the same data. The detector's AGPL licence is one script away from a permissive one.

IMPACT



Faults found as they develop, not at the next survey

Aerial surveys are usually yearly, so faults sit for weeks to months. This watches every array all the time.



A deadline in place of an alarm

B-17 must be acted on before 14:00. After that the damage is projected to stop being recoverable.



Every loss priced, with its working

3.07 MWh over 72 h is ₹7,509 at ₹2.446/kWh: the two SECI Bhadla Phase-III lots, blended by capacity.



One fault, left for a month

₹74,114: 1.01 MWh/day × 30 days. An illustration that assumes a constant loss, not a measurement.

BENEFITS



Operational: the crew knows what to do first

A ranked list with reasons, and a day plan for two crews. No guessing which alarm matters.



Economic: more energy from capacity already built

Loss stated in rupees at the plant's own tariff, and no drone flown where imaging adds nothing.



Environmental: more clean energy delivered

A plant that loses less to faults delivers more of what it was built for.

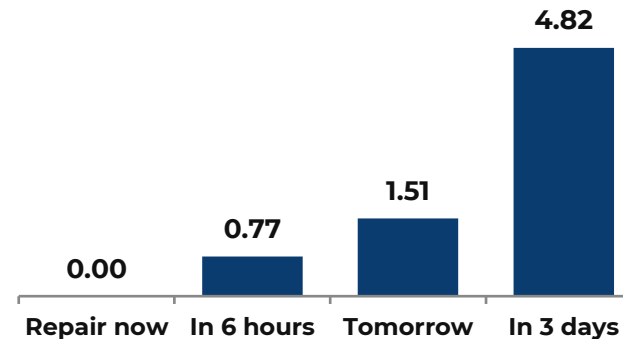


Social: safer field work

The planner keeps crews out of the field in the hours above 40 °C.

THE COST OF WAITING: ARRAY B-17

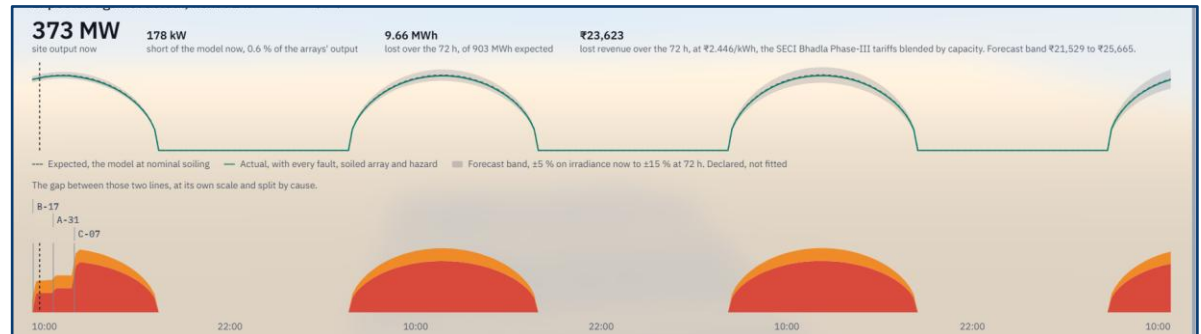
Energy lost if the repair starts then
(MWh)



Fix it now and nothing more is lost. Wait three days and it is 4.82 MWh, ₹11,795.

From the B-17 incident screen, as captured on 7 Oct 2026: ₹1,894 at 6 hours, ₹3,704 tomorrow. All three delays run past the 14:00 deadline. The figures move with site time.

A defect report cannot give this.



Analytics, in the prototype: expected against actual for the modelled arrays over 72 h, and the shortfall by cause. No deviation settlement charge is computed: rupees are lost energy times the tariff, and nothing else.

FUTURE SCOPE



Research: finish the thermal classifier and report its held-out metric per class. Fit the forecast band to real misses. Radiometric thermal data, and field validation of the deadline model.



Patent: none filed. Candidate for a novelty search: a defect, its mechanism and a forecast turned into a repair deadline and a crew plan that re-derives live.



Startup: software for O&M contractors and owners of utility-scale plants, fed by the plant's own SCADA. Needs a permissively licensed detector first.

REFERENCES

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2. NREL. PVWatts Version 5 Manual, NREL/TP-6A20-60272. docs.nrel.gov/docs/fy14osti/60272.pdf
3. SECI auction, Bhadla Phase-III Solar Park, 2017. iea.org/policies/6373-auction-of-solar-corporation-of-india-seci; pv-magazine-india.com/?p=1613
4. Raptor Maps. InfraredSolarModules. github.com/RaptorMaps/InfraredSolarModules
5. Solar Panel Fault Detection v2, Roboflow Universe. universe.roboflow.com/solarvision-gwljt/solar-panel-fault-detection
6. CERC Deviation Settlement Mechanism Regulations, 2024: context only. No charge is computed from it.

WHAT WE BUILT ON

Kind	What	Licence
Dataset	Solar Panel Fault Detection v2, Roboflow: 921 images	CC BY 4.0
Dataset	InfraredSolarModules, Raptor Maps: 20,000 images	MIT
Pre-trained model	Ultralytics YOLOv8n, fine-tuned by us	AGPL-3.0
External API	Groq, model openai/gpt-oss-120b	Groq terms
Libraries	Next.js, React, three, React Three Fiber, drei, zustand, zod, Tailwind	MIT
Libraries	onnxruntime-web, highs (HiGHS in WebAssembly)	MIT
Libraries	postprocessing; lucide-react; playwright-core, TypeScript	Zlib; ISC; Apache-2.0
Typeface	IBM Plex, in the product	OFL



Reference we owe: the loop follows the RAISE-winning Robinsun solar agent. We rebuilt it with a trained defect model and a physics-grounded simulation in place of a physical drone. No personal or proprietary data is used; telemetry is simulated.



Team SIGMOID: Rehaan Ahmad Khan, Shantanu Singh, Lakshita Rawat, Krishna Agarwal. Code: github.com/RAK2315/solar-proj (AGPL-3.0)